

Two-Step Equations & Checking Solutions

Answer Key

Grade 6–7 Mathematics

Quick Answers

- | | |
|-------|-------------------------|
| 1. 4 | 7. (a) = 17, — |
| 2. 7 | 8. 8 |
| 3. 20 | 9. 7 |
| 4. -3 | 10. -4 |
| 5. 18 | 11. (a) = 6, (b) = 6, — |
| 6. 3 | 12. 5 |
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What is verified

10 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 7, 11. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 6–7 Mathematics

7.EE.B.4 – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Difficulty 1–4 of 5 across 15 tagged checks.

Common wrong answers

11. *If they got 3: isolated x by subtracting 6 from both sides instead of adding it*

1. Solve and check: $2x + 3 = 11$

Solution: Subtract 3 from both sides: $2x = 8$. Divide by 2: $x = 4$. Check: $2(4) + 3 = 11$.

✓

$x = 4$

2. Solve and check: $3x - 5 = 16$

Solution: Add 5 to both sides: $3x = 21$. Divide by 3: $x = 7$. Check: $3(7) - 5 = 16$.

✓

$x = 7$

3. Solve and check: $\frac{x}{4} + 2 = 7$

Solution: Subtract 2 from both sides: $\frac{x}{4} = 5$. Multiply by 4: $x = 20$. Check: $20 \div 4 + 2 =$

7. ✓

$x = 20$

4. Solve and check: $5x + 9 = -6$

Solution: Subtract 9 from both sides: $5x = -15$. Divide by 5: $x = -3$. Check: $5(-3) + 9 = -6$. ✓

$$x = -3$$

5. Solve and check: $\frac{x}{3} - 8 = -2$

Solution: Add 8 to both sides: $\frac{x}{3} = 6$. Multiply by 3: $x = 18$. Check: $18 \div 3 - 8 = -2$. ✓

$$x = 18$$

6. Solve and check: $10 - 2x = 4$

Solution: Subtract 10 from both sides: $-2x = -6$. Divide by -2 (dividing by a negative keeps the equation balanced): $x = 3$. Check: $10 - 2(3) = 4$. ✓

$$x = 3$$

7. Nina says $x = 6$ solves $4x - 7 = 17$. (a) evaluate $4x - 7$ at $x = 6$; (b) is she correct?

Solution: (a) $4(6) - 7 = 24 - 7 = 17$.

$$17$$

(b) Model answer: yes — the left side comes out to 17, exactly the right side of the equation, so $x = 6$ makes the equation true and is the solution. (Instructor judges the sentence; it must connect the matching value to “is a solution.”)

8. Solve and check: $7 = 2x - 9$

Solution: The variable is on the right — the steps are the same. Add 9 to both sides: $16 = 2x$. Divide by 2: $x = 8$. Check: $2(8) - 9 = 7$. ✓

$$x = 8$$

9. Solve and check: $\frac{x+5}{3} = 4$

Solution: The *whole* sum $x + 5$ is divided by 3, so multiply both sides by 3 first: $x + 5 = 12$. Subtract 5: $x = 7$. Check: $(7 + 5) \div 3 = 4$. ✓

$$x = 7$$

10. Solve and check: $-4x + 3 = 19$

Solution: Subtract 3 from both sides: $-4x = 16$. Divide by -4 : $x = -4$. Check: $-4(-4) + 3 = 16 + 3 = 19$. ✓

$$x = -4$$

11. Leo solves $4x - 6 = 18$ and gets $x = 3$. (a) check his answer; (b) solve correctly; (c) name his error.

Solution: (a) $4(3) - 6 = 12 - 6 = 6$, not 18 — so $x = 3$ fails the check.

$$6$$

(b) Add 6 to both sides: $4x = 24$. Divide by 4: $x = 6$. Check: $4(6) - 6 = 18$. ✓

$$x = 6$$

(c) Model answer: Leo *subtracted* 6 from both sides instead of adding it — to undo -6 you add 6, so he computed $(18 - 6) \div 4 = 3$ instead of $(18 + 6) \div 4$. (Instructor judges the sentence.)

12. **Challenge.** Bike rental: flat fee 4 dollars plus 2.50 dollars per hour, total 16.50 dollars: solve $2.5h + 4 = 16.5$.

Solution: Subtract 4 from both sides: $2.5h = 12.5$. Divide by 2.5: $h = 5$. Check: $2.5(5) + 4 = 12.5 + 4 = 16.5$. ✓ So Amara rented for 5 hours.

$$h = 5$$